AI ENGINEERING SERIES

100 Hands-On AI Projects

From Beginner to Research-Grade

100 Real-World Projects · Build · Learn · Ship

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100 Hands-On AI Projects

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FOREWORD

This book exists because the best way to learn AI is to build things with it.

100 Hands-On AI Projects is a practical field guide — not a textbook. It does not ask you to read before you build. Each chapter describes a real, runnable project, explains why it matters, shows you the tools and steps needed to replicate it, and points you to a real GitHub repository to explore. The goal is to get you building faster, and to keep you building longer.

How This Book Is Organised

The 100 projects are divided into three progressive levels:

Beginner (Chapters 1–30) Projects that can be completed with basic Python knowledge, minimal setup, and free or low-cost APIs. No ML theory required — just follow the steps.

Intermediate (Chapters 31–60) Projects for developers comfortable with Python and some ML concepts. You'll work with real datasets, real APIs, and real-world deployment challenges.

Advanced (Chapters 61–100) Research-grade and production-scale projects. These require GPU access, deep ML knowledge, and patience — but the results are state-of-the-art systems you can be proud of.

How to Use Each Chapter

Every chapter follows a consistent structure: a plain-English overview explaining what the project is and why it matters; a summary card showing difficulty and estimated time; the tools and stack; step-by-step execution guide; a Pro Tip from hard-won experience; key concepts to cement your learning; ideas for taking the project further; and a GitHub link to a working reference implementation.

Now open a terminal, pick a project, and start building.

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SECTION I

Beginner Projects

Chapters 1–30

These thirty projects are designed for anyone with basic Python knowledge. You do not need prior machine learning experience — just the willingness to follow along, run some code, and see what happens. By the end of this section you will have built real working systems with NLP, computer vision, and recommendation engines, and you will understand the core tools that the rest of the book builds on.

Chapters in this section are organised to build on each other. Each project stands alone, but they reward sequential reading.

"The measure of intelligence is the ability to change."

— Albert Einstein

"In God we trust. All others must bring data."

— W. Edwards Deming

"The only way to learn a new programming language is by writing programs in it. The same is true for AI."

— Dennis Ritchie, adapted

CHAPTER 01

Chatbot with Personality

Beginner Difficulty: ●○○○○ ⏱ Estimated time: **2–3 hrs**

KEY CONCEPTS

- **LLM**
- **System Prompt**
- **Chat Loop**

OVERVIEW

Build a conversational AI chatbot powered by the OpenAI API that responds in a chosen persona --- a pirate, a friendly tutor, or a sci-fi robot. This is the classic first AI project for a reason: it teaches you how LLMs work, how to structure prompts, and how to maintain a conversation loop.

This foundational project teaches you how LLMs work from the inside out — how system prompts shape model behaviour, how to maintain stateful conversations, and how quickly the modern AI tooling stack gets you to a working product. It is the classic first AI project for good reason.

WHAT YOU'LL BUILD

- A terminal-based or web chat interface
- Custom system prompts that define personality
- Conversation history that keeps context across messages

TOOLS & STACK

```
Python, OpenAI API (GPT-3.5-turbo), Flask or Streamlit (optional for
UI)
```

STEPS TO EXECUTE

- Install dependencies: pip install openai
- Get your OpenAI API key from platform.openai.com and store it in a .env file
- Write a system prompt that defines the persona (e.g., \'You are a friendly pirate who only speaks in nautical metaphors\')